

Hochiminh city University of Technology
Faculty of Computer Science and Engineering

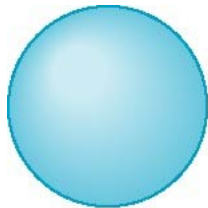


COMPUTER GRAPHICS

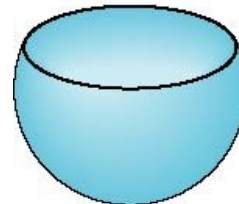
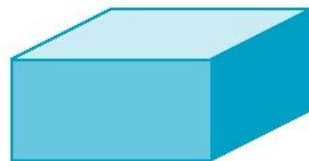
CHAPTER 3:

Front and Back Faces

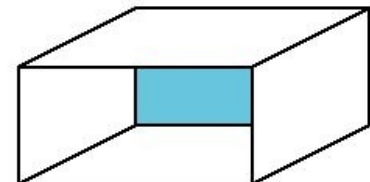
- ❑ The default is shade only front faces which works correctly for convex objects
- ❑ If we set two sided lighting, OpenGL will shade both sides of a surface
- ❑ Each side can have its own properties which are set by using **GL_FRONT**, **GL_BACK**, or **GL_FRONT_AND_BACK**



back faces not visible



back faces visible



Front and Back Faces

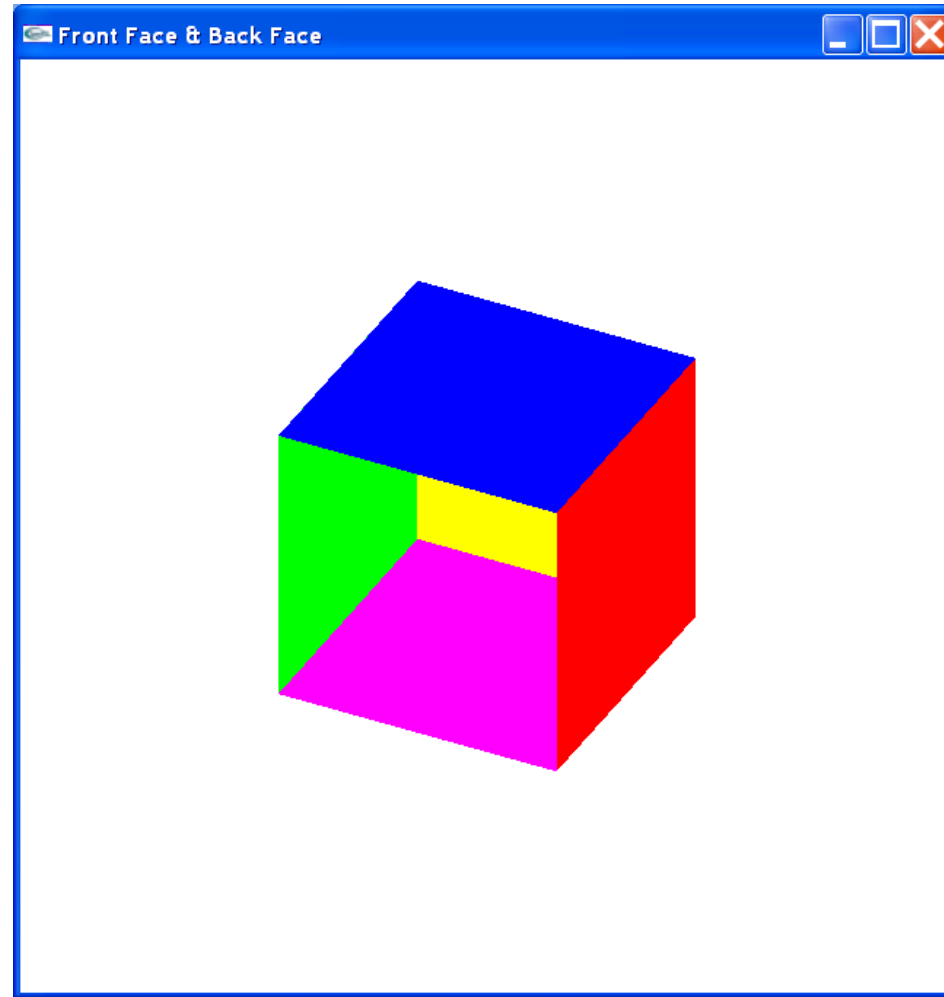
❑ Specify Front Faces

- `glFrontFace(GL_CCW)`, `glFrontFace(GL_CW)`

❑ Cull Face

- `glEnable(GL_CULL_FACE)`
- `glCullFace(GLenum mode);`
 - `GL_FRONT`,
 - `GL_BACK`,
 - `GL_FRONT_AND_BACK`

Front and Back Faces



Front and Back Faces

```
glBegin(GL_QUADS);
    glColor3f(1, 0, 0); // x = 1
    glVertex3f(1, 1, 1);
    glVertex3f(1, -1, -1);
    glColor3f(0, 1, 0); //x = -1
    glVertex3f(-1, 1, 1);
    glVertex3f(-1, -1, -1);
    glColor3f(0, 0, 1); // y = 1
    glVertex3f( 1, 1, 1);
    glVertex3f(-1, 1, -1);
    glColor3f(1, 0, 1); // y = -1
    glVertex3f( 1, -1, 1);
    glVertex3f(-1, -1, -1);
    ..... // z = -1
glEnd();
```

```
    glVertex3f(1, -1, 1);
    glVertex3f(1, 1, -1);

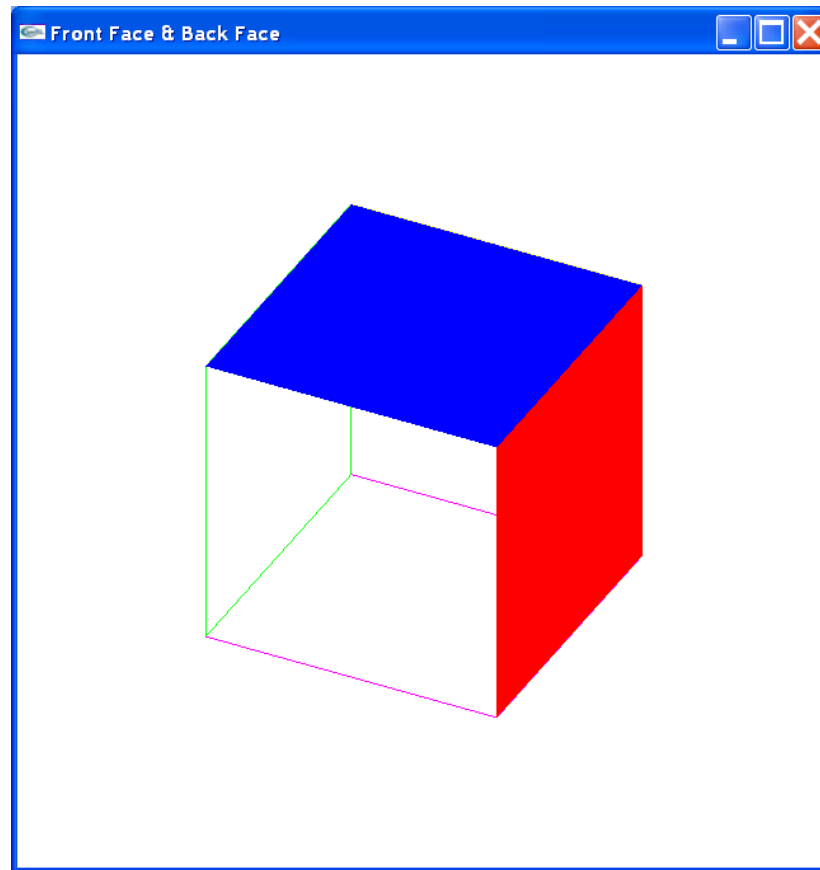
    glVertex3f(-1, 1, -1);
    glVertex3f(-1, -1, 1);

    glVertex3f( 1, 1, -1);
    glVertex3f(-1, 1, 1);

    glVertex3f(-1, -1, 1);
    glVertex3f( 1, -1, -1);
```

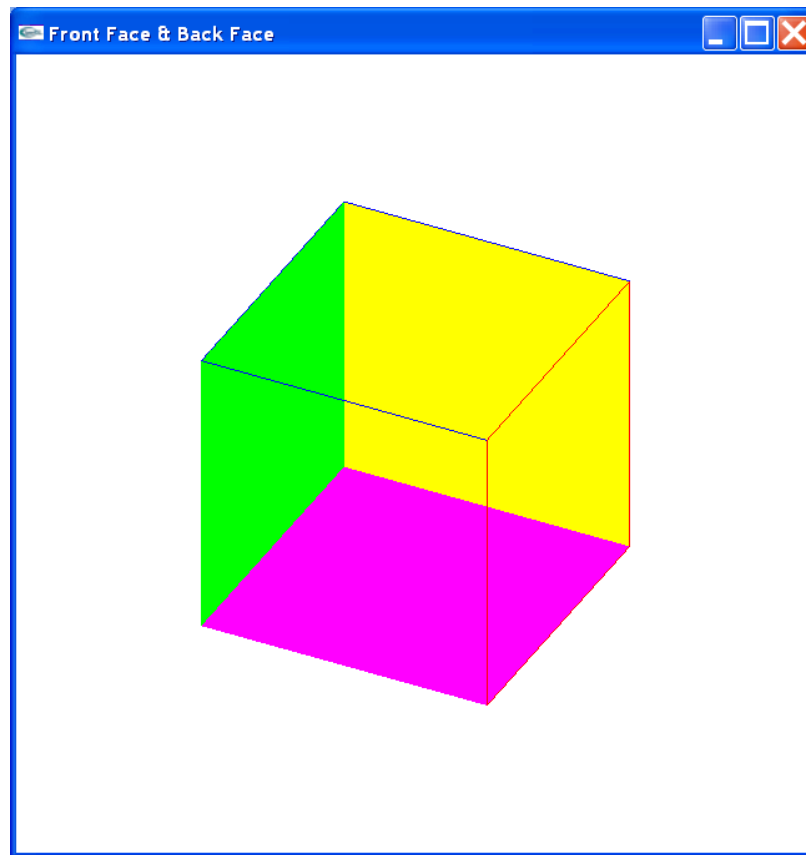
Front and Back Faces

```
glPolygonMode(GL_FRONT, GL_FILL);  
glPolygonMode(GL_BACK, GL_LINE);
```



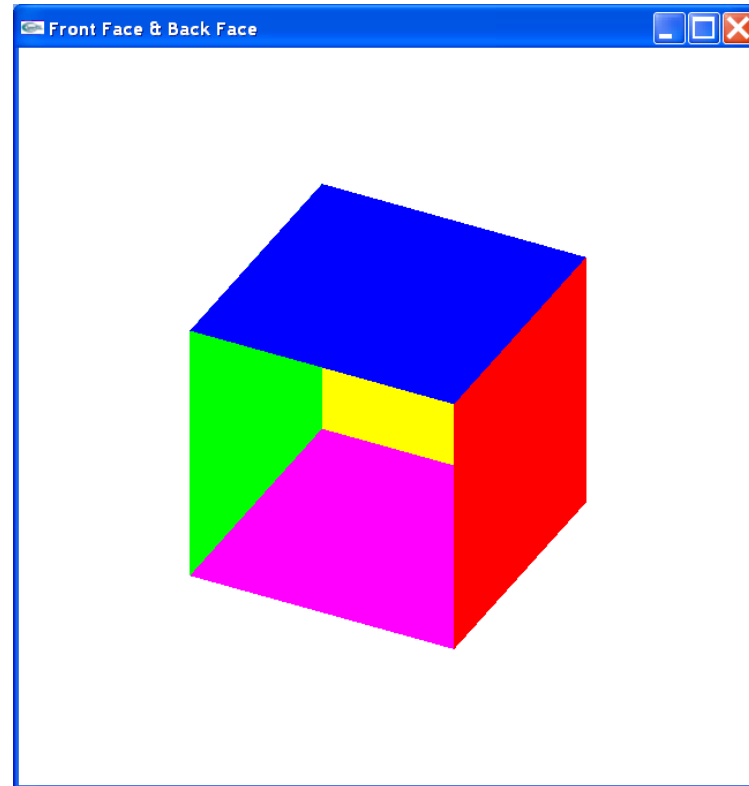
Front and Back Faces

```
glFrontFace(GL_CW);  
glPolygonMode(GL_FRONT, GL_FILL);  
glPolygonMode(GL_BACK, GL_LINE);
```



Front and Back Faces

```
glFrontFace(GL_CCW); //default  
glPolygonMode(GL_FRONT, GL_FILL);  
glPolygonMode(GL_BACK, GL_FILL);
```



Front and Back Faces

```
glFrontFace(GL_CCW);  
glEnable(GL_CULL_FACE);  
glCullFace(GL_BACK);
```

